gat-typst Package Manual

Version 0.1.0

A comprehensive Typst package for academic writing, homework assignments, and cheatsheets

Contents

1 Introduction

The `gat-typst` package is a comprehensive template system for academic writing in Typst, with a focus on mathematics-heavy homework assignments, cheatsheets, and technical documents. It provides opinionated defaults, extensive math utilities, and a curated collection of the best packages from the Typst ecosystem.

This is a growing library for Gatlen Culp's personal usage and is unstable.

1.1 Key Features

- **Templates**: Ready-to-use templates for homework, cheatsheets, and general documents
- **Math Utilities**: Enhanced math notation with semantic symbols and custom functions
- **Integrated Packages**: 20+ carefully selected packages for diagrams, code, and formatting
- **Opinionated Styling**: Professional defaults for code, tables, and equations
- **Convenience Functions**: Colored boxes, custom math operators, and more

1.2 Installation

To use this package locally, ensure it's installed in your local Typst packages directory:

Shell
```shell
# Unix-like systems
~/.local/share/typst/packages/local/gat-typst/0.1.0/

# Windows
%APPDATA%/typst/packages/local/gat-typst/0.1.0/
```

Then import it in your Typst documents:

Typst
```typst
#import "@local/gat-typst:0.1.0": *
```

1.3 Quick Start

Here's a minimal homework document to get started:

Typst
```typst
#import "@local/gat-typst:0.1.0": *

#show: homework.with(
  title: "Homework 1",
  course: "CS 101",
  university: "Your University",
  author: "Your Name",
  email: "you@example.com",
  date: datetime.today(),
)

#question(status: "done")[
  Prove that $P = N P$.
]

#solution[
```

```
  Left as an exercise to the reader.

  Using our custom math utilities:
  $Ex(X^2) >= (Ex(X))^2$ by Jensen's inequality.
]
```

2 Important Behavior Changes

When using this package, several default behaviors are modified for improved mathematical typesetting and consistency.

2.1 Math Shorthands

The package enables extensive math operator shorthands via the `quick-maths` package. These are **automatically active** in all templates.

2.1.1 Complete Shorthand List

Operators:

```typst
$a ** b$   // Small asterisk
$a * b$    // Centered dot ⚠
$a +- b$   // Plus-minus
$A |- B$   // Turnstile
$a o+ b$   // Circled plus
$a o. b$   // Circled dot
$a @ b$    // Also circled dot
```

$a * b \quad a \cdot b \quad a \pm b$

$A \vdash B \quad a \oplus b$

$a \odot b \quad a \odot b$

Logic:

```typst
$a .. b$          // Spacing (quad)
$A :. B$          // Therefore
$A :' B$          // Because
```

$a \quad b$

$A \therefore B \qquad A \because B$

Relations:

```typst
$a -<= b$  // Precedes or equals
$a -< b$   // Precedes
$a >-= b$  // Succeeds or equals
$a >- b$   // Succeeds
$a -= b$   // Equivalent
$a ~= b$   // Asymptotically equal
$a ~~ b$   // Approximately
```

$a \preceq b \quad a \prec b$

$a \succeq b \quad a \succ b$

$a \equiv b \quad a \cong b \quad a \approx b$

Brackets:

```typst
$<< a, b >>$      // Angle brackets
$a == b$          // Equals (unchanged)
```

$\langle a, b \rangle$

$a = b$

2.1.2 Warning About * Shorthand

The `*` $\rightarrow$ `dot.op` replacement is the most impactful change:

```typst
// With shorthands (default):
$a * b$             // Renders: a · b
$a ** b$            // Renders: a * b (small asterisk)

// To use literal asterisk:
$a ast b$           // Renders: a * b
```

$a \cdot b$ (renders as $a \cdot b$)

$a * b$ (small asterisk)

$a \cdot b$ (literal asterisk)

This follows conventions in linear algebra and physics but may surprise users expecting traditional asterisk multiplication.

2.1.3 Disabling Shorthands

If you need to disable shorthands:

```typst
#show: common.with(enable-shorthands: false)
```

Note: Currently this parameter exists but shorthands are applied regardless. This will be fixed in a future version.

2.1.4 Examples

```typst
// Statistics
$Ex(X) .. Var(X) = Ex(X^2) - (Ex(X))^2$

// Linear algebra
$A * B != B * A .. "in general"$

// Logic
$x > 0 :. x^2 > 0$

// Approximation
$e ~~ 2.718$

// Relations
$x -<= y "iff" f(x) <= f(y)$
```

$\mathbb{E}[X] \quad \mathrm{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

$A \cdot B \neq B \cdot A \quad \text{in general}$

$x > 0 \therefore x^2 > 0$

$e \approx 2.718$

$x \preceq y \text{ iff } f(x) \leq f(y)$

2.2 Superscript Transformations

Three special superscript transformations are enabled for common mathematical notation.

2.2.1 Transpose (^T)

```typst
$A^T = "transpose of" A$
$x^T y = iprod(x, y)$
$(A B)^T = B^T A^T$
```

$A^\top = \text{transpose of } A$

$x^\top y = \langle x, y \rangle$

$(AB)^\top = B^\top A^\top$

Renders with proper transpose symbol ($^\top$) instead of superscript T.

2.2.2 Dagger (^+)

```typst
$A^+ = "Moore-Penrose pseudoinverse"$
$H^+ = "Hermitian transpose"$
```

$A^\dagger = \text{Moore-Penrose pseudoinverse}$

$H^\dagger = \text{Hermitian transpose}$

Renders as dagger symbol (†).

2.2.3 Star (^*)

```typst
$V^* = "optimal value function"$
$pi^* = "optimal policy"$
$x^* = arg min_x f(x)$
```

$V^\star = \text{optimal value function}$

$\pi^\star = \text{optimal policy}$

$x^\star = \arg\min_x f(x)$

Renders with star symbol ($\star$) instead of superscript asterisk.

This distinguishes from the asterisk operator and provides cleaner mathematical notation.

2.2.4 Interaction with Shorthands

The superscript transformations work correctly even when $*$ shorthand is enabled:

```typst
$A^T * B^*$  // Transpose A times optimal B
```

$A^\top \cdot B^\star$

3 Utilities

The package provides extensive utilities for math, boxes, and display formatting.

3.1 Math Utilities

3.1.1 Statistics Functions

Colored, bracketed notation for statistical operators:

```typst
$Ex(X)$          // Expected value
$Var(X)$         // Variance
$Cov(X, Y)$      // Covariance
$Corr(X, Y)$     // Correlation
$Pr(X > 0)$      // Probability
```

$\mathbb{E}[X]$ $\quad$ $\mathrm{Var}[X]$ $\quad$ $\mathrm{Cov}[X]$

$\mathrm{Corr}[X]$ $\quad$ $\Pr[X > 0]$

These render with blue coloring and square brackets. Example:

$\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$ by Jensen's inequality.

$\mathrm{Var}[X + Y] = \mathrm{Var}[X] + \mathrm{Var}[Y]$ when X and Y are independent.

3.1.2 Reasoning Operators

```typst
$x st x > 0$          // "such that"
$If x > 0 Then y = 1 Else y = 0$
```

$x \text{ s.t. } x > 0$

$\textbf{if } x > 0 \textbf{ then } y = 1 \textbf{ else } y = 0$

Example: $\max\{x \text{ s.t. } x \in \mathbb{R}\}$

3.1.3 Machine Learning / RL

```typst
$KL(p, q)$       // KL divergence
$refPol$         // Reference policy
```

$D_{\mathrm{KL}}(p \parallel q)$ $\qquad$ $\overline{\pi}$

Example: Minimize $D_{\mathrm{KL}}(\pi \parallel \overline{\pi})$ to stay close to reference policy.

3.1.4 Miscellaneous

```typst
$SE$             // Standard error
$ddot$           // Double dot
$alias$          // Alias arrow
$clamp(x)$       // Clamp function
$solve(x)$       // Solve function
```

SE $\quad$ $\ddot{}$ $\quad$ $\xleftarrow{\text{alias}}$

$\mathrm{clamp}\{x\}$ $\quad$ $\mathrm{solve}\{x\}$

3.1.5 Custom Math Function Factory

Create your own math functions with custom delimiters:

```typst
#let myFunc = math-func("myFunc", delim: "[")
$myFunc(x, y)$

// Or with custom left/right delimiters
#let norm = math-func("norm", ldelim: math.bar.v.double, rdelim: math.bar.v.double)
$norm(x)$
```

myFunc$[x]$

norm$\|x\|$

3.2 Box Utilities

3.2.1 `gatbox()`

Customizable colored boxes with automatic color cycling.

3.2.1.1 Basic Usage

```typst
#gatbox[
  This is a basic box with default black color.
]

#gatbox(color: blue)[
  This is a blue box.
]
```

This is a basic box with default black color.

This is a blue box.

3.2.1.2 Color Cycling

Automatically cycle through predefined colors:

```typst
#gatbox(color-cycle: true)[First box - blue]
#gatbox(color-cycle: true)[Second box - red]
#gatbox(color-cycle: true)[Third box - green]
#gatbox(color-cycle: true)[Fourth box - orange]
#gatbox(color-cycle: true)[Fifth box - purple]
```

First box - blue

Second box - red

Third box - green

Fourth box - orange

Fifth box - purple

3.2.1.3 Parameters

- `color` (color, default: `black`): Box color
- `color-cycle` (boolean, default: `false`): Auto-cycle through colors
- `title` (content): Optional title
- All `showybox` parameters supported

3.2.1.4 Examples

```typst
#gatbox(title: "Important", color: red)[
  This is a critical point!
]

#gatbox(color-cycle: true, title: "Theorem 1")[
  All horses are the same color.
]
```

Important

This is a critical point!

Theorem 1

All horses are the same color.

3.2.2 `showybox`

The base box system (re-exported from the `showybox` package) with `breakable: true` by default.

```
#showybox(
  title: "Note",
  frame: (
    border-color: blue,
    title-color: blue.lighten(80%),
  ),
)[
  Content here can break across pages.
]
```

Note

Content here can break across pages.

3.3 Display Utilities

3.3.1 Show Rules

The package includes custom show rules that can be used independently:

```
#show: super-ast-as-star

$V^* = V(x^*)$  // Asterisk renders as star symbol
```

$V^\star = V(x^\star)$

4 Common Styling

4.1 Code Styling

All code blocks are automatically styled:

4.1.1 Automatic Changes

1. **Font size**: Reduced to 8pt for compactness
2. **Zebra striping**: Alternating row backgrounds (gray.lighten(80%))
3. **Border**: 1pt black stroke around code blocks
4. **Syntax highlighting**: Language-specific coloring

4.1.2 Example

Python

```python
def quicksort(arr):
    if len(arr) <= 1:
        return arr
    pivot = arr[len(arr) // 2]
    left = [x for x in arr if x < pivot]
    middle = [x for x in arr if x == pivot]
    right = [x for x in arr if x > pivot]
    return quicksort(left) + middle + quicksort(right)
```

No manual configuration needed - just use standard Typst raw blocks with language tags.

4.2 Table Styling

The `booktabs` package is automatically applied, giving all tables professional styling with:

- Proper spacing between rows
- Professional horizontal rules (top, mid, bottom)
- No vertical lines (following booktabs philosophy)
- Better visual hierarchy

Typst

```typst
#table(
  columns: 3,
  [*Algorithm*], [*Time*], [*Space*],
  [Quicksort], [$O(n log n)$], [$O(log n)$],
  [Mergesort], [$O(n log n)$], [$O(n)$],
  [Heapsort], [$O(n log n)$], [$O(1)$],
)
```

No special syntax required - standard `#table()` gets enhanced styling.

5 Templates

The package provides three main templates: `common`, `homework`, and `cheatsheet`. All templates are designed to work together and share common styling.

5.1 Common Template

The `common` template is the foundation that provides imports, styling, and configuration. It's automatically applied by both `homework` and `cheatsheet` templates, but can also be used standalone.

5.1.1 Usage

```typst
#show: common.with()
```

5.1.2 Parameters

- `enable-shorthands` (boolean, default: `true`): Enable math operator shorthands

5.1.3 Features Provided

The `common` template automatically configures:

1. **FontAwesome Icons**: Access to icon library via `fa-icon()` function
2. **Code Styling**: Syntax highlighting with Codly, 8pt font size, zebra striping
3. **Table Styling**: Professional booktabs-style tables by default
4. **Math Configuration**:
   - Equation numbering with `equate` package
   - Breakable equations
   - Sub-numbering disabled
5. **Show Rules**:
   - `^T` renders as transpose
   - `^+` renders as dagger (†)
   - `^*` renders as star ($\star$) instead of asterisk

5.1.4 Example

```typst
#import "@local/gat-typst:0.1.0": *

#show: common.with()

// Use FontAwesome icons
#fa-icon("heart") Math is beautiful

// Code with automatic styling
```

python def hello(): print("Hello, world!")

```

 // Professional tables
 #table(
   columns: 3,
   [*Name*], [*Age*], [*City*],
   [Alice], [25], [Boston],
```

```
  [Bob], [30], [Seattle],
)

// Math with special superscripts
$ A^T dot B^+ = C^* $
```

5.2 Homework Template

The `homework` template creates professional homework assignments with custom components for questions, solutions, and more.

5.2.1 Usage

```typst
#show: homework.with(
  title: "Homework 1",
  course: "Course Name",
  university: "University Name",
  author: "Your Name",
  email: "your@email.com",
  collaborators: [List of collaborators],
  resources: [Resources used],
  date: datetime.today(),
  cols: 1,  // or 2 for two-column layout
  show-header: true,
  paper-size: "us-letter",
)
```

5.2.2 Parameters

- `title` (string): Homework title (e.g., "HW01", "Problem Set 3")
- `course` (content): Course name or code
- `university` (content): University name
- `author` (content): Your name
- `email` (content): Your email address
- `collaborators` (content, default: `[]`): List of collaborators
- `resources` (content, default: `[]`): Resources used
- `date` (datetime, default: `datetime.today()`): Assignment date
- `cols` (integer, default: `1`): Number of columns (1 or 2)
- `show-header` (boolean, default: `true`): Show header with metadata
- `paper-size` (string, default: `"us-letter"`): Paper size

5.2.3 Components

5.2.3.1 `question()`

Creates a colored box for problem statements with optional status tracking.

```typst
#question(status: "done")[
  What is the derivative of $x^2$?
]

#question(status: "partial")[
  This is partially complete.
]

#question(status: "todo")[
```

```
  Not started yet.
]

#question()[
  No status indicator.
]
```

Status colors:

- `"done"`: Green
- `"partial"`: Yellow
- `"todo"`: Red
- `"none"` or omitted: Black

5.2.3.2 `solution()`

Wrapper for solution content with optional column layout.

```typst
#solution[
  The derivative is $2x$.
]

#solution(cols: 2)[
  This solution spans two columns.
]
```

5.2.3.3 `boxed-solution`

Inline box for short solutions.

```typst
The answer is #boxed-solution[$x = 5$].
```

5.2.3.4 `sub-solution`

Nested solution box for multi-part problems.

```typst
#sub-solution[
  For part (a), we first note that...
]
```

5.2.3.5 `notes`

Purple box for additional notes or remarks.

```typst
#notes[
  Remember to check edge cases!
]
```

5.2.4 Complete Example

```typst
#import "@local/gat-typst:0.1.0": *

#show: homework.with(
```

```
  title: "HW02",
  course: "Linear Algebra",
  university: "MIT",
  author: "Student Name",
  email: "student@mit.edu",
  collaborators: [Alice, Bob],
  resources: [Textbook Chapter 3],
  date: datetime(year: 2025, month: 1, day: 15),
  cols: 1,
)

= Matrix Operations

#question(status: "done")[
  Let $A = mat(1, 2; 3, 4)$. Compute $A^T A$.
]

#solution[
  We compute:
  $
    A^T A = mat(1, 3; 2, 4) mat(1, 2; 3, 4)
    = mat(10, 14; 14, 20)
  $

  #sub-solution[
    Note that $A^T A$ is always symmetric.
  ]
]
```

5.3 Cheatsheet Template

The `cheatsheet` template creates compact, three-column cheatsheets perfect for exams.

5.3.1 Usage

```typst
#show: cheatsheet.with(
  title: "Midterm",
  course: "Course Name",
  university: "University Name",
  author: "Your Name",
  email: "your@email.com",
  date: datetime.today(),
)
```

5.3.2 Parameters

Similar to `homework` but without `cols`, `show-header`, and `paper-size` (fixed to 3-column layout on US Letter with 0.5cm margins).

5.3.3 Features

- **3-column layout** with 0.3cm gutter
- **Compact spacing**: 6pt base font, 7pt level-1 headings, 6pt level-2/3
- **No page numbers**: Maximizes space
- **Centered headings**: Better visual organization

5.3.4 Example

```typst
#import "@local/gat-typst:0.1.0": *

#show: cheatsheet.with(
  title: "Final Exam Cheatsheet",
  course: "Calculus I",
  author: "Your Name",
  date: datetime(year: 2025, month: 5, day: 10),
)

= Derivatives

== Power Rule
$ dv(, x) x^n = n x^(n-1) $

== Product Rule
$ dv(, x) (f g) = f' g + f g' $

== Chain Rule
$ dv(, x) f(g(x)) = f'(g(x)) g'(x) $

= Integrals

```

```
== Power Rule
$ integral x^n dd(x) = x^(n+1)/(n+1) + C $
```

6 Dependencies & Imports

The package integrates 20+ external packages, organized by category.

6.1 Diagramming

6.1.1 CeTZ (v0.4.2)

Comprehensive drawing library for creating custom diagrams.

```typst
#import "@preview/cetz:0.4.2": canvas, draw

#canvas({
  import draw: *
  circle((0, 0), radius: 1)
  line((0, 0), (1, 1))
  content((0, 0), $x$)
})
```

Use for: Custom diagrams, network architectures, geometric figures.

6.1.2 CeTZ Plot (v0.1.2)

Plotting extension for CeTZ.

```typst
#import "@preview/cetz-plot:0.1.2": plot

#canvas({
  plot.plot(
    size: (8, 6),
    x-tick-step: 1,
    y-tick-step: 1,
    {
      plot.add(x => x * x, domain: (-2, 2))
    }
  )
})
```

Use for: Function plots, data visualization.

6.1.3 Fletcher (v0.5.8)

Arrow diagrams and state machines.

```typst
#diagram(
  node((0, 0), $A$, radius: 2em),
  node((1, 0), $B$, radius: 2em),
  edge((0, 0), (1, 0), "->", $f$),
)
```

Use for: State machines, flowcharts, commutative diagrams.

6.1.4 Lilaq (v0.5.0)

Modern data visualization library.

```typst
#lq.diagram(
  title: [Data Plot],
  xlabel: $x$,
  ylabel: $y$,
  lq.plot((0, 1, 2, 3), (1, 4, 2, 3), mark: "o"),
)
```

Use for: Scientific plots, cleaner syntax than CeTZ Plot.

6.2 Math

6.2.1 Matset (v0.1.0)

Set notation support.

Use for: Mathematical set operations and notation.

6.2.2 Equate (v0.3.2)

Advanced equation numbering with breakable equations.

```typst
$
  f(x) = x^2
$ <eq:quadratic>

Reference: @eq:quadratic
```

$$f(x) = x^2$$

Use for: Equation numbering and cross-referencing. Automatically enabled.

6.2.3 Physica (v0.9.5)

Comprehensive physics notation package.

Key functions:

```typst
$vb(x)$                 // Vector bold
$vu(u)$                 // Unit vector
$dd(x)$                 // Differential
$dv(f, x)$              // Derivative
$pdv(f, x, y)$          // Partial
$grad f$, $div vb(F)$, $curl vb(F)$
$iprod(u, v)$           // Inner product
$norm(x)$               // Norm
$abs(x)$                // Absolute value
$Set(x, x > 0)$         // Set builder
$dmat(1, 2, 3)$         // Diagonal matrix
$evaluated(f)_a^b$      // Evaluation bar
```

$\boldsymbol{x}$ $\quad \hat{\boldsymbol{u}}$ $\quad \mathrm{d}x$

$\frac{\mathrm{d}f}{\mathrm{d}x}$ $\quad \frac{\partial^2 f}{\partial x \partial y}$

$\boldsymbol{\nabla} f$ $\quad \boldsymbol{\nabla} \cdot \boldsymbol{F}$ $\quad \boldsymbol{\nabla} \times \boldsymbol{F}$

$\langle u, v \rangle$ $\quad \|x\|$ $\quad |x|$

$\{x \mid x > 0\}$

$$\begin{pmatrix} 1 & & \\ & 2 & \\ & & 3 \end{pmatrix}$$

$f|_a^b$

Example:

```
$
  dv(vb(x), t) = vb(v), quad
  pdv(f, x, y, [2, 1]) = 0, quad
  iprod(vb(u), vb(v)) = norm(vb(u)) norm(vb(v)) cos theta
$
```

$$\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} = \boldsymbol{v}, \quad \frac{\partial^3 f}{\partial x^2 \partial y} = 0, \quad \langle \boldsymbol{u}, \boldsymbol{v} \rangle = \|\boldsymbol{u}\| \|\boldsymbol{v}\| \cos\theta$$

Use for: Almost all physics and advanced math notation.

6.2.4 Quick-maths (v0.2.1)

Math shorthand system (powers the operator replacements).

Automatically enabled via `common` template. See section 5 for details.

6.2.5 Mannot (v0.3.0)

Math annotations and labels.

TODO: FIX

```
$
  p_i = exp(-beta E_i) / sum_j exp(-beta E_j)
  #annot(<beta>, pos: top)[Inverse temperature]
$
```

Use for: Annotating complex equations, teaching materials.

6.3 Code

6.3.1 Callisto (v0.2.4)

Jupyter notebook rendering.

```
#let (render, Cell, In, Out) = callisto.config(
  nb: json("notebook.ipynb"),
)

#In("cell-tag")   // Show input cell
#Out("cell-tag")  // Show output cell
#Cell("cell-tag") // Show both
```

Use for: Including Jupyter notebook code/output in documents.

6.3.2 Lovelace (v0.3.0)

Pseudocode formatting.

```
#pseudocode-list[
  + *Input:* Array $A$ of length $n$
  + *Output:* Sorted array
  + *for* $i = 1$ *to* $n$ *do*
    + Process element $A[i]$
```

```
]
```

Use for: Algorithm descriptions.

6.3.3 Codly (v1.3.0) + Codly Languages (v0.1.1)

Modern syntax highlighting for code blocks.

Automatically configured with:

- 8pt font size
- 1pt black stroke
- Zebra striping (gray.lighten(80%))

```python
def fibonacci(n):
    if n <= 1:
        return n
    return fibonacci(n-1) + fibonacci(n-2)
```

Supports: Python, Rust, JavaScript, TypeScript, Java, C++, and 50+ more languages.

Use for: All code snippets. Automatically applied to raw blocks.

6.3.4 Cmarker (v0.1.6)

CommonMark (Markdown) support.

Use for: Converting Markdown to Typst.

6.4 Display & Formatting

6.4.1 Frame-it (v1.2.0)

Custom frame styles for content boxes.

```typst
#let (example, definition) = frames(
  example: ("Example", blue),
  definition: ("Definition", green),
)

#example[Title][Tag][
  Body content here.
]
```

Use for: Theorem environments, examples, definitions.

6.4.2 FontAwesome (v0.6.0)

Font Awesome icon support (version 6).

```typst
#fa-icon("heart")
#fa-icon("calculator")
#fa-icon("robot")
```

Use for: Visual indicators, status icons. Note: Requires Font Awesome fonts installed.

6.4.3 Showybox (v2.0.4)

Styled, breakable boxes (base for `gatbox`).

```typst
#showybox(
  title: "Title",
  frame: (border-color: blue),
  breakable: true,
)[
  Content
]
```

Use for: Custom boxes, callouts, highlighted content.

6.4.4 Hydra (v0.6.2)

Context-aware page headers showing current section.

```typst
#set page(header: context [
  #text(fill: gray)[*#hydra(2)*]
])
```

Use for: Displaying current section in headers. Automatically enabled in `homework` template.

6.4.5 Booktabs (v0.0.4)

Publication-quality table styling.

Automatically applied to all tables. Creates professional spacing and rules.

Use for: All tables. No manual configuration needed.

6.5 Compatibility

6.5.1 MiTeX (v0.2.5)

LaTeX compatibility layer for including LaTeX code.

```typst
#mitex(`
  \begin{bmatrix}
    1 & 2 \\
    3 & 4
  \end{bmatrix}
`)
```

Use for: Converting existing LaTeX, complex LaTeX-only constructs.

6.6 Color

6.6.1 Splash (v0.5.0)

Extended color support including xcolor names.

```typst
#text(fill: xcolor("red!50!blue"))[Mixed color]
```

Use for: Advanced color specifications.

6.6.2 Typpuccino (v0.1.0)

Catppuccin color schemes (frappe, latte, macchiato, mocha).

```typst
#import "@preview/typpuccino:0.1.0": mocha

#text(fill: mocha.red)[Catppuccin red]
```

Use for: Consistent color theming.

7 Complete Examples

7.1 Simple Homework Document

```typst
#import "@local/gat-typst:0.1.0": *

#show: homework.with(
  title: "Problem Set 1",
  course: "Calculus II",
  university: "Your University",
  author: "Your Name",
  email: "you@example.com",
  date: datetime(year: 2025, month: 2, day: 1),
)

= Integration by Parts

#question(status: "done")[
  Evaluate $integral x e^x dd(x)$.
]

#solution[
  Using integration by parts with $u = x$ and $dv(, x) = e^x dd(x)$:

  $
    integral x e^x dd(x)
    &= x e^x - integral e^x dd(x) \
    &= x e^x - e^x + C \
    &= e^x (x - 1) + C
  $
]
```

7.2 Math-Heavy Solution with Custom Functions

```typst
#import "@local/gat-typst:0.1.0": *

#show: homework.with(
  title: "Statistics Assignment",
  course: "Probability Theory",
  author: "Student",
  email: "student@school.edu",
)

= Random Variables

#question(status: "done")[
  Let $X$ and $Y$ be independent random variables.
  Show that $Var(X + Y) = Var(X) + Var(Y)$.
```

```
]

#solution[
  By definition:
  $
    Var(X + Y) &= Ex((X + Y)^2) - (Ex(X + Y))^2 \
    &= Ex(X^2 + 2X Y + Y^2) - (Ex(X) + Ex(Y))^2 \
    &= Ex(X^2) + 2Ex(X Y) + Ex(Y^2) - (Ex(X))^2 - 2Ex(X)Ex(Y) - (Ex(Y))^2
  $

  Since $X$ and $Y$ are independent, $Ex(X Y) = Ex(X) * Ex(Y)$:
  $
    Var(X + Y) &= Ex(X^2) - (Ex(X))^2 + Ex(Y^2) - (Ex(Y))^2 \
    &= Var(X) + Var(Y)
  $

  #sub-solution[
    *Note:* This result generalizes: for independent $X_1, ..., X_n$,
    $
      Var(sum_(i=1)^n X_i) = sum_(i=1)^n Var(X_i)
    $
  ]
]
```

7.3 Compact Cheatsheet

```
Typst
#import "@local/gat-typst:0.1.0": *

#show: cheatsheet.with(
  title: "Midterm Cheatsheet",
  course: "Linear Algebra",
  author: "Your Name",
  date: datetime(year: 2025, month: 3, day: 15),
)

#let gatbox = gatbox.with(color-cycle: true)

= Matrix Operations

#gatbox(title: [Transpose])[
  $(A^T)_(i j) = A_(j i)$

  $(A B)^T = B^T A^T$
]

#gatbox(title: [Determinant])[
  $det(A B) = det(A) * det(B)$
```

```
  $det(A^T) = det(A)$

  $det(A^(-1)) = 1/det(A)$
]

= Eigenvalues

#gatbox(title: [Definition])[
  $A vb(v) = lambda vb(v)$

  $det(A - lambda I) = 0$
]

#gatbox(title: [Properties])[
  $product_(i=1)^n lambda_i = det(A)$

  $sum_(i=1)^n lambda_i = "tr"(A)$
]

= Vector Spaces

#gatbox(title: [Subspaces])[
  *Null space:* $cal(N)(A) = Set(vb(x), A vb(x) = vb(0))$

  *Column space:* $cal(C)(A) = Set(A vb(x), vb(x) in RR^n)$

  *Row space:* $cal(C)(A^T)$
]

#gatbox(title: [Fundamental Theorem])[
  $dim cal(C)(A) + dim cal(N)(A) = n$

  $cal(C)(A)^perp = cal(N)(A^T)$
]
```

8 Tips & Best Practices

Based on conventions and optimal usage of this package:

8.1 Math Notation

1. **Use semantic symbols** instead of generic ones:

```typst
// Good
$grad f$              // Gradient
$product_(i=1)^n x_i$ // Product

// Avoid
$nabla f$
$Pi_(i=1)^n x_i$
```

2. **Space before subscripts with functions:**

```typst
// Good
$V_pi (s)$   // Value function of policy π at state s
$Q_theta (s, a)$

// Bad (no space before parenthesis)
$V_pi(s)$
$Q_theta(s, a)$
```

3. **Use physica utilities for common notation:**

```typst
$vb(x)$          // Vector bold
$vu(n)$          // Unit vector
$cal(A)$         // Mathcal letters (script)
$dd(x)$          // Differential
$dv(f, x)$       // Derivative
$pdv(f, x, y)$   // Partial derivative
$iprod(u, v)$    // Inner product (not angle brackets)
$norm(x)$        // Norm (not ||x||)
$abs(x)$         // Absolute value
$eval(f)_a^b$    // Evaluation bar
```

4. **Leverage custom statistics functions:**

```typst
// These are colored and use square brackets
$Ex(X)$      // Expectation
$Var(X)$     // Variance
$Pr(A)$      // Probability
```

8.2 Layout

1. **Two-column homework** for dense problems:

```typst
#show: homework.with(
```

```typst
  cols: 2,
  // ... other params
)

// Override for specific solutions
#solution(cols: 1)[
  This needs full width.
]
```

2. **Color-cycle boxes** for organizing content:

```typst
#gatbox(color-cycle: true, title: "Theorem 1")[...]
#gatbox(color-cycle: true, title: "Theorem 2")[...]
#gatbox(color-cycle: true, title: "Theorem 3")[...]
// Automatically cycles: blue → red → green → orange → purple
```

3. **Status tracking** for homework progress:

```typst
#question(status: "done")[Completed problem]
#question(status: "partial")[Work in progress]
#question(status: "todo")[Not started]
```

8.3 Organization

1. **Use sub-solutions** for complex multi-part problems:

```typst
#solution[
  Main solution approach...

  #sub-solution[
    For part (a), we have...
  ]

  #sub-solution[
    For part (b), note that...
  ]
]
```

2. **Add notes** for clarifications or caveats:

```typst
#notes[
  This assumes the matrix is invertible.
  For singular matrices, use pseudoinverse.
]
```

3. **Label equations** for cross-referencing:

```typst
$
  f(x) = x^2 + 2x + 1
$ <eq:quadratic>
```

```

From @eq:quadratic, we see that...
```

8.4 Code

1. **Always specify language** for proper highlighting:

```python
def example():
    pass
```

Not just:

```
def example():
    pass
```

2. **Use Callisto for notebook integration:**

```typst
#let (render, Cell, In, Out) = callisto.config(
  nb: json("notebook.ipynb"),
)

// Tag cells in notebook, then reference them
#In("data-loading")
#Out("plot-results")
```

8.5 Performance

1. **Compile time**: Heavy imports may slow compilation. Consider splitting large documents.

2. **Font Awesome**: Requires fonts installed system-wide. If icons don't render, install Font Awesome fonts.

3. **Cheatsheet margins**: 0.5cm is very tight. Adjust if printing issues occur:

```typst
#set page(margin: 0.7cm)  // Before cheatsheet.with()
```

8.6 Common Pitfalls

1. **Shorthand conflicts**: If you need literal `*` for multiplication:

```typst
$a ast b$  // Use 'ast' explicitly
```

2. **Superscript confusion**: Remember `^*` becomes star:

```typst
$x^*$      // Renders as x⋆ (star)
$x^ast$    // Renders as x* (asterisk if needed)
```

3. **Column overrides**: Don't forget to override `solution` for full-width content in 2-column layouts:

```typst
#let solution = solution.with(cols: 1)
```

9 Reference

9.1 Quick Import Guide

```typst
// Full import
#import "@local/gat-typst:0.1.0": *

// Selective import
#import "@local/gat-typst:0.1.0": homework, question, solution, gatbox

// Import specific utilities
#import "@local/gat-typst:0.1.0": Ex, Var, KL, vb, cal
```

9.2 Template Comparison

Feature	common	homework	cheatsheet
Columns	N/A	1 or 2	3 (fixed)
Font size	11pt	13pt (1-col) / 10pt (2-col)	6pt
Headers	None	Optional	None
Margins	Default	1 inch	0.5cm
Question boxes	No	Yes	No (manual)
Equation numbers	Yes	Yes	Optional

9.3 Function Reference

9.3.1 Templates

- `common(enable-shorthands: true, body)`
- `homework(title, course, university, author, email, ...)`
- `cheatsheet(title, course, university, author, email, ...)`

9.3.2 Components

- `question(status: "none", ...)`
- `solution(cols: 1, body)`
- `boxed-solution`
- `sub-solution(...)`
- `notes(...)`

9.3.3 Boxes

- `gatbox(color: black, color-cycle: false, ...)`
- `showybox(...)`

9.3.4 Math (Statistics)

- `Ex()` - Expectation
- `Var()` - Variance
- `Cov()` - Covariance
- `Corr()` - Correlation

- `Pr()` - Probability

9.3.5 Math (Other)

- `KL(p, q)` - KL divergence
- `refPol` - Reference policy
- `st`, `If`, `Else`, `Then` - Logic
- `clamp(x)`, `solve(x)` - Functions

9.3.6 Utilities

- `date-format(date)` - Format datetime
- `math-func(name, ...)` - Create custom math function
- `super-ast-as-star` - Show rule for `^*`

9.4 Further Reading

For detailed package documentation:

- **Physica**: https://github.com/Leedehai/typst-physics
- **CeTZ**: https://cetz-package.github.io/docs/
- **Fletcher**: https://github.com/typst/packages/blob/main/packages/preview/fletcher/
- **Codly**: https://typst.app/universe/package/codly
- **Typst Documentation**: https://typst.app/docs/

9.5 Version History

v0.1.0 (Current)
- Initial release
- Templates: homework, cheatsheet, common
- 20+ integrated packages
- Math utilities and shorthands
- Custom box system

—

For issues, suggestions, or contributions, see the project repository.